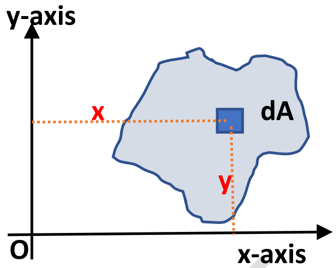
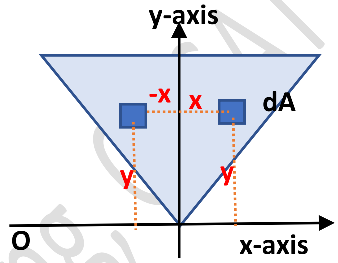
Product of inertia

Product of inertia of a plane area with respect to x-y axes is defined by $I_{xy} = \int x\, y\, dA$ where an elemental area is multiplied by the product of its distances from the x and y axes and integrated over the entire area.

Unlike moment of inertia, the product of inertia can be zero or negative.



Consider any shape having an axis of symmetry. Here, y-axis is an axis of symmetry. If we are finding out the product of inertia, we will consider an elemental area dA at distance (x,y) from the axes, find (xy dA) and integrate over the entire region. Since the y-axis is a symmetrical axis, for any element dA that we consider on the right side of the axis at distance x, there is a corresponding element at distance (-x) on the left side of the symmetrical axis.



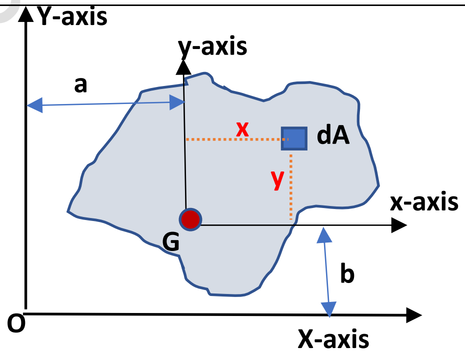
So when we add up (xy dA) on the right side with (-xy dA) on the left, we get zero. That is the product of inertia of an area with respect to a symmetrical axis and another axis perpendicular to it is always zero. The axes about which the product of inertia becomes zero are known as principal axes.

Parallel axis theorem for product of inertia

Consider a plane area whose centroid is at G. With respect to the co-ordinate axes X-Y passing through O, the centroid is at (a, b). Let us fix another co-ordinate axes x-y at point G. With respect to this centroidal axes, the centroid is at (0, 0). That is,

$(\bar{X}, \bar{Y}) = (a, b)$

$(\bar{x}, \bar{y}) = (0,0)$



Consider an elemental area dA at distances x, y from the centroidal axes. Product of inertia of the area with respect to X-Y axes is

$$I_{XY} = \int XY\, dA = \int (a+x)(b+y) dA = \int xy\, dA + \int ab\, dA + \int a\, y\, dA + \int b\, x\, dA$$
$$= \int xy\, dA + ab \int dA + a \int y dA + b \int x\, dA$$

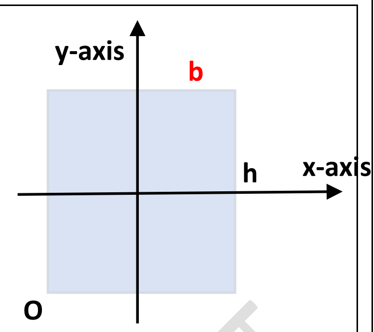
From the topic of centroid, we know that $\frac{\int x\, dA}{\int dA} = \bar{x}$ and $\frac{\int y\, dA}{\int dA} = \bar{y}$, where $(\bar{x}, \bar{y})$ is the cordinates of the centroid of the area with repect to the axis from which the distance x and y are measured. Here (x,y) is measured from the axes passing through G. Therefore, $\bar{x} = 0, \bar{y} = 0$. Hence $\int ydA = 0$

$I_{XY} = \int xy\, dA + ab \int dA = \overline{I_{xy}} + (ab)\, A$ where $\overline{I_{xy}}$ is the product of inertia of the area with respect to the its centroid axes.

This is known as parallel axis theorem for product of inertia which states that product of inertia of an area with respect to any X-Y coordinate system is equal to the sum of (the product of inertia of the area with respect to a parallel centroidal x-y coordinate system) and (the product of the area and the distances between the parallel coordinate systems).

Example 1. Determine the product of inertia of a rectangular area with respect to its centroidal x-y axes

Since the rectangular area is symmetrical with respect to its centroidal x-y axes, $\bar{I}_{xy} = 0$

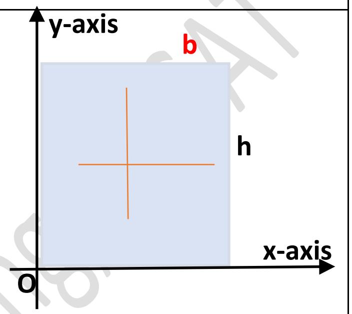


Example 2. Determine the product of inertia of a rectangular area with respect to the x-y axes passing through the edges

By parallel axis theorem, $I_{XY} = \bar{I}_{xy} + (ab)\,A$

But $\bar{I}_{xy} = 0$, the distances between the parallel coordinate systems are $(\frac{b}{2}, \frac{h}{2})$.

$$I_{XY} = 0 + \left(\frac{b}{2} \times \frac{h}{2}\right)(bh) = \frac{b^2h^2}{4}$$

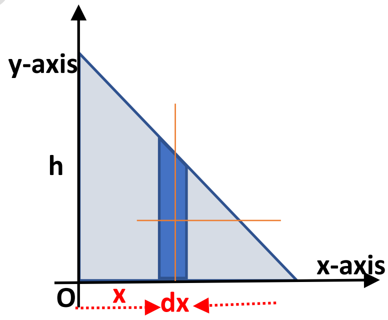


Example 3. Determine the product of inertia of a triangular area with respect to the x-y axes passing through the edges

Consider an infinitesimally small rectangular element of width dx and height y at a distance x from the y axis. Area of the infinitesimally small element, $dA = y\,dx$.

From the similar triangles, we can write $\frac{h}{y} = \frac{b}{b-x}$ >>> $y = \frac{h}{b}(b-x)$

$dA = \frac{h}{b}(b-x)\,dx$



If we can find out the product of inertia of the elemental area with respect to the x-y axes, then the product of inertia of the triangle will be the integral of that.

By parallel axis theorem, the product of inertia of the elemental area about the x-y axes= product of inertia of the elemental area about its centroidal axes + Product of the elemental area and the distances between the parallel set of axes.

$$dI_{XY} = d\bar{I}_{xy} + (ab)\,dA$$

product of inertia of the elemental rectangular area about its centroidal axes, $d\bar{I}_{xy} = 0$ (since the element is symmetrical with respect to its centroidal axes)

The distance between the set of axes (a,b) = $(x, \frac{y}{2})$

$$dI_{XY} = 0 + \left(x.\frac{y}{2}\right)(y\,dx) = x.\frac{h}{b}\frac{(b-x)}{2}\,\frac{h}{b}(b-x)\,dx = \frac{h^2}{2b^2}x(b-x)^2\,dx$$

$$I_{xy} = \int_0^b \left(\frac{h^2}{2b^2}x(b-x)^2\right)dx = \frac{h^2}{2b^2}\int_0^b x(b^2 - 2bx + x^2)dx = \frac{h^2}{2b^2}\left[\frac{b^2x^2}{2} - \frac{2bx^3}{3} + \frac{x^4}{4}\right]_0^b = \frac{b^2h^2}{24}$$

(The integration is w.r.t x, the limits for x varies from 0 to b)

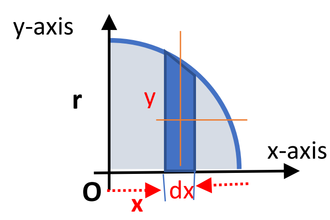
#Exercise 22

22.1

Using parallel axis theorem, determine the product of inertia of the right triangular area with respect to its centroidal x and y axes.

Example 4. Determine the product of inertia of a circular area with respect to the x-y axes shown.

Consider an infinitesimally small rectangular element of width dx and height y at a distance x from the y axis. Area of the infinitesimally small element, $dA = y\,dx$.



From the equation of a circle $x^2 + y^2 = r^2$, we can write $y = \sqrt{r^2 - x^2}$

$dA = \sqrt{r^2 - x^2}\,dx$

By parallel axis theorem, the product of inertia of the elemental area about the x-y axes= product of inertia of the elemental area about its centroidal axes + product of the elemental area and the distances between the parallel set of axes.

$$dI_{XY} = d\bar{I}_{xy} + (ab)\,dA$$

Product of inertia of the elemental rectangular area about its centroidal axes, $d\bar{I}_{xy} = 0$ (since the element is symmetrical with respect to its centroidal axes)

The distance between the set of axes (a, b) = $(x, \frac{y}{2})$

$$dI_{XY} = 0 + \left(x.\frac{y}{2}\right)(y\,dx) = x.\frac{\sqrt{r^2-x^2}}{2}\ \sqrt{r^2 - x^2}\,dx = \frac{x}{2}(r^2 - x^2)\,dx$$

$$I_{xy} = \int_0^r \left(\frac{x}{2}(r^2 - x^2)\right) dx = \left[\frac{r^2x^2}{4} - \frac{x^4}{8}\right]_0^r = \frac{r^4}{8}$$

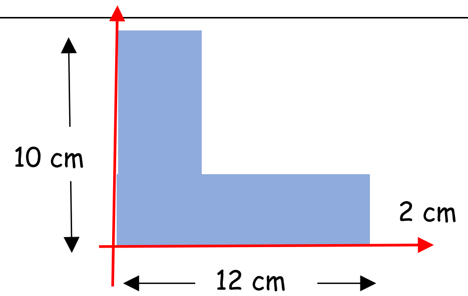
(The integration is w.r.t x, the limits for x varies from 0 to r)

#Exercise 23

23.1

Using parallel axis theorem, determine the product of inertia of the quarter circular area with respect to its centroidal x and y axes.

Example 5. Calculate the product of inertia of the angle section shown in the figure with respect to the axes shown.



Let us divide the given shape in to two rectangles as shown.

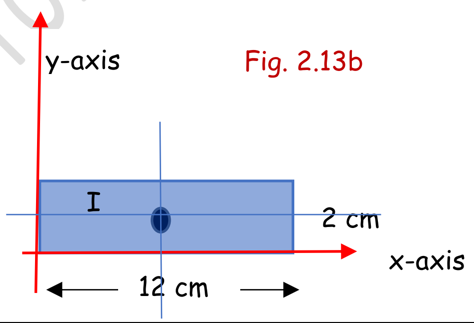


Fig. 2.13b

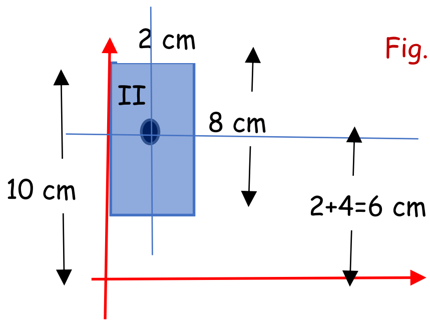


Fig. 2.13c

Product of inertia of rectangle I with respect to the given x-y axes, $I_{1xy} = \frac{b^2h^2}{4} = \frac{12^2 \times 2^2}{4} = 144\ cm^4$

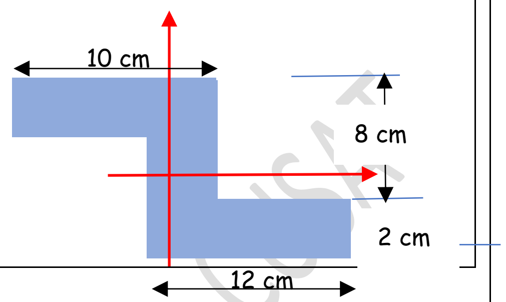
Since the x axis does not pass the bottom edge of II, we need to apply the parallel axis theorem.

P.I. of rectangle II with respect to the given x-y axes = P.I of rectangle II with respect to its centroidal axes + (area of rectangle x product of the distances between the two sets of axes)

$$I_{2xy} = \overline{I_{2xy}} + A_2\ (ab) = 0 + (2 \times 8)(1 \times 6) = 96\ cm^4$$

Therefore, $I_{xy} = I_{1xy} + I_{2xy} = 144 + 96 = 240\ cm^4$

Example 6 Calculate the product of inertia of the area with respect to the x-y axes passing through its centroid. The width of the section is 2cm throughout.



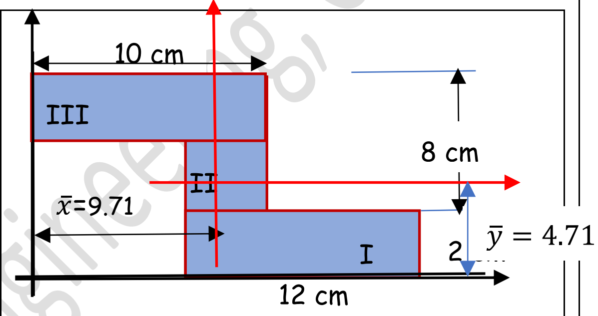
Let us locate the centroid of the given shape with respect to any co-ordinate system. Consider a co-ordinate system as shown passing through the left and bottom ends.

Divide the given shape in to three rectangles as shown

A_1=12×2=24cm^2	x_1= 8+6 =14cm	y_1=1 cm
A_2=6×2=12cm^2	x_2= 8+1 =9cm	y_2=2+3=5 cm
A_3=10×2=20cm^2	x_3= 5cm	y_3=10-1=9 cm

$$\bar{x} = \frac{A_1x_1 + A_2x_2 + A_3x_3}{A_1 + A_2 + A_3} = 9.71\ cm$$

$$\bar{y} = \frac{A_1y_1 + A_2y_2 + A_3y_3}{A_1 + A_2 + A_3} = 4.71\ cm$$



The centroid of the given shape is at (9.71cm, 4.71cm) from the left and bottom edges. Now let us fix the co-ordinate system at the centroid, with respect to which the product of inertia is to be found out. They are marked in red. We need to apply the parallel axes theorem to find the PI of the rectangle 1 w.r.t the x-y axes

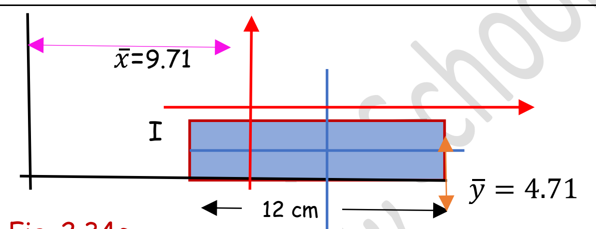


Fig. 2.34c

$$I_{1xy} = \overline{I_{1xy}} + A_1\ (y_1 - \bar{y})(x_1 - \bar{x})$$

$$= 0 + (24)(1 - 4.71)(14 - 9.71) = -382\ cm^4$$

(The Centroid of the rectangle I is in the fourth quadrant with respect to the given x-y axes in red, So, the x-distance should be +ve and y-distance -ve)

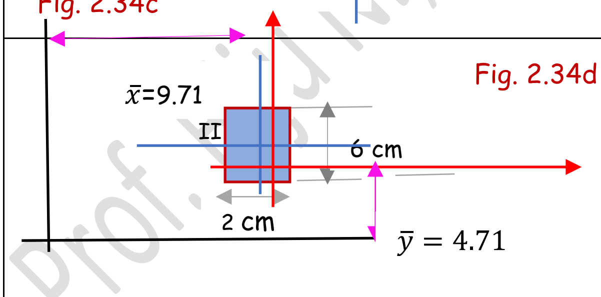


Fig. 2.34d

$$I_{2xy} = \overline{I_{2xy}} + A_2\ (y_2 - \bar{y})(x_2 - \bar{x})$$

$$= 0 + (12)(5 - 4.71)(9 - 9.71) = -2.47\ cm^4$$

(The Centroid of the rectangle II is in the second quadrant with respect to the given x-y axes in red, So, the x-distance is -ve and y-distance is +ve)

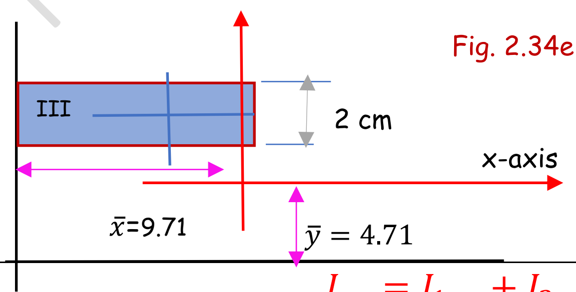


Fig. 2.34e

$$I_{3xy} = \overline{I_{3xy}} + A_3\ (y_3 - \bar{y})(x_3 - \bar{x})$$

$$= 0 + (20)(9 - 4.71)(5 - 9.71) = -404.12\ cm^4$$

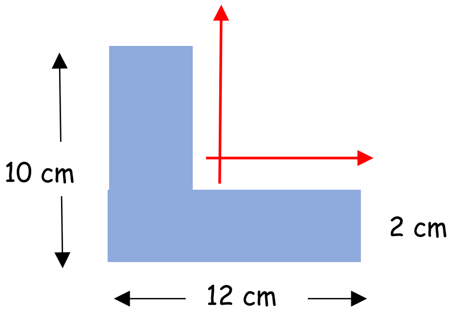
(The Centroid of the rectangle III is in the second quadrant.So, the x-distance is -ve and y-distance +ve)

$$I_{xy} = I_{1xy} + I_{2xy} + I_{3xy} = -788.59\ cm^4$$

#Exercise 24

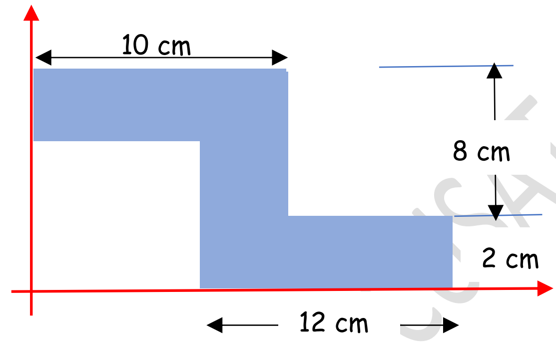
24. 1

Calculate the product of inertia of the given shape with respect to its centroidal x-y axes.



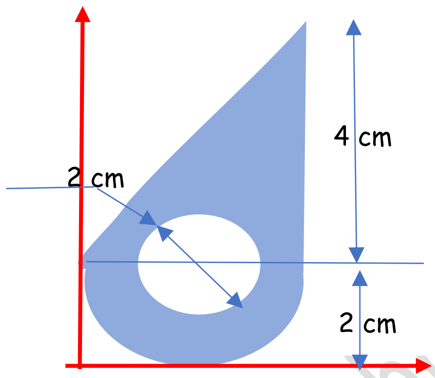
24. 2

Calculate the product of inertia of the given shape with respect to the given x-y axes.



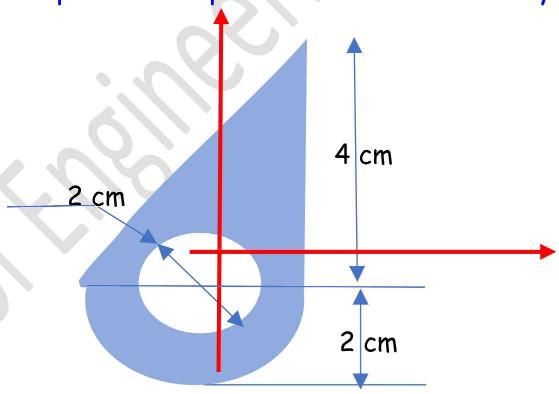
24. 3

Calculate the product of inertia of the given shape with respect to its x-y axes.



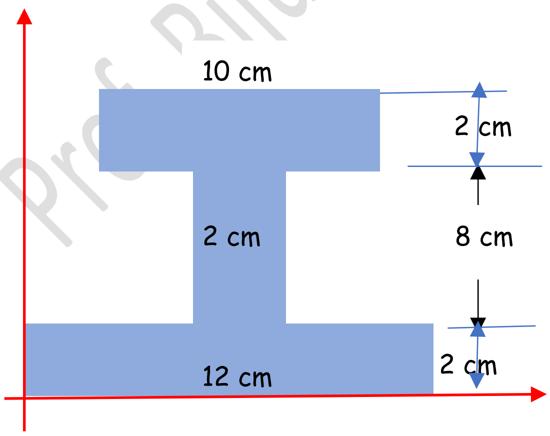
24. 4

Calculate the product of inertia of the given shape with respect to its centroidal x-y axes



24. 5

Calculate the product of inertia of the I section with respect to the x-y axis. What will be its PI with respect to its centroidal x-y axes



24. 6

what is the relation between r and h so that x and y axes become the principal axes.